EC2 & Web hosting using Windows instance

Class no:13

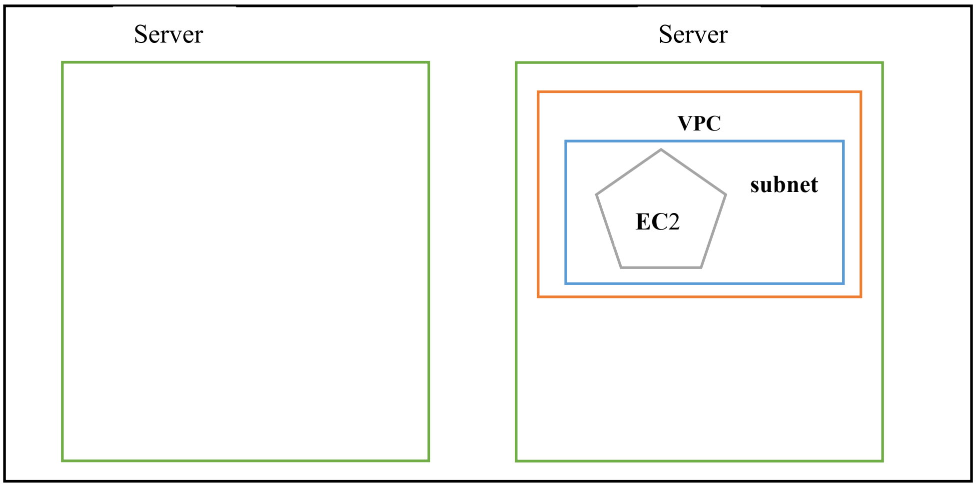
EC2:

Amazon EC2 (Elastic Compute Cloud) is AWS's core service for renting virtual servers in the cloud. To configure EC2, you need to create an instance by choosing an AMI (Amazon Machine Image), instance type, storage, networking, and security settings. EC2 instance families are divided into categories like General Purpose, Compute Optimized, Memory Optimized, Storage Optimized, and Accelerated Computing, each balancing performance and price differently.

To configure an instance, we need:

- Name
- OS
- Instance type -> CPU & Memory
- Key pair (Public and Private key)
- Disk Storage (Persistence Volume)
- Network settings
  - ✓ VPC (Virtual Private Cloud)-larger network
  - ✓ Subnet (smaller network)
  - ✓ Auto-assign public Ip (Enabled: public ec2/ Disabled: private ec2)
  - ✓ Security groups
  - ✓ Disk Storage (Persistence Volume)

Layout of an Ec2 instance:



INSTANCE TYPE

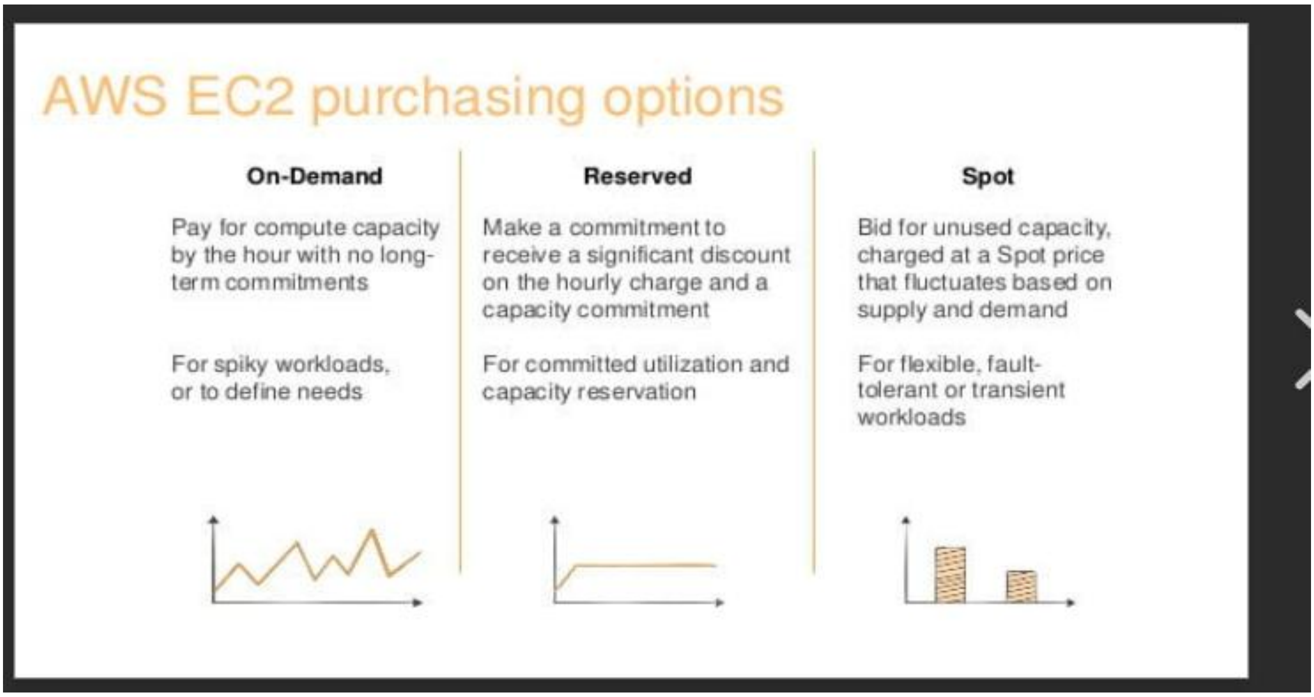
Amazon EC2 instance types are predefined virtual server configurations that combine varying amounts of compute, memory, storage, and networking capacity to suit different workloads. Choosing the right type ensures optimal performance and cost-efficiency.

Main Categories: (Performance based)

- **General Purpose** – Balanced compute, memory, and networking. Ideal for web servers, code repositories, and small-to-medium databases. Examples: T / M Series
- **Compute Optimized** – High-performance processors for compute-bound tasks like batch processing, media transcoding, and game servers. Examples: C series
- **Memory Optimized** – High memory-to-CPU ratio for large in-memory datasets, analytics, and enterprise apps. Examples: R / X / Z Series
- **Accelerated Computing** – Use GPUs or FPGAs for graphics rendering, ML training, or scientific simulations. Examples: P5, G6e, Inf2, Trn1.
- **Storage Optimized** – High sequential read/write and IOPS for large datasets. Examples: I / D /H Series.
- **High-Performance Computing** (HPC) – Optimized for large-scale simulations and deep learning. Examples: Hpc7g, Hpc6a.

Purchasing or Pricing models

- On demand: On-Demand Instances provide **predictable pricing** and are billed per second with no long-term commitments. These instances are ideal for workloads that require **reliable and uninterrupted performance.**
- Spot instance: Spot Instances allow you to utilize unused EC2 capacity at significantly reduced costs, often up to **90% cheaper** than On-Demand pricing. However, these instances come with the risk of **interruption** when AWS reclaims capacity for On-Demand or Reserved Instances. Spot Instances are ideal for workloads that are **flexible** and can tolerate interruptions.
- Reserved instance: Reserved Instances (RIs) are a **billing discount model** offered by AWS for services like Amazon EC2 and Amazon RDS. They allow users to save significantly—up to **75% compared to On-Demand pricing**—by committing to a specific instance configuration for a one-year or three-year term. Unlike physical instances, RIs are not actual servers but rather a **pricing mechanism** applied to On-Demand Instances that match specific attributes.



KEY PAIR:

- key pair is a set of public and private keys used for secure login to your EC2 instances.
- AWS uses public-key cryptography:

  (i)The public key is stored in AWS and attached to your EC2 instance.

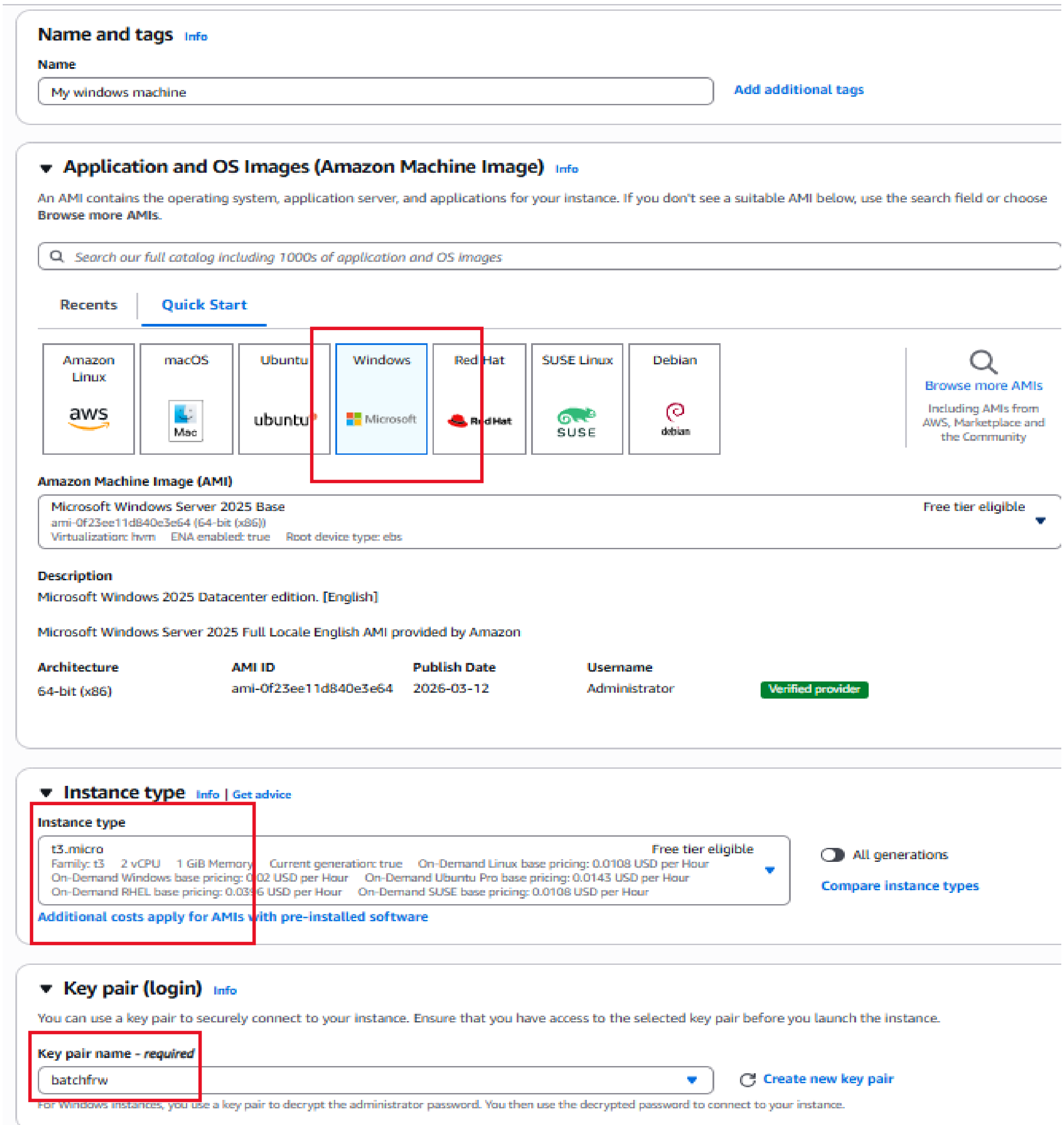
  (ii)The private key is downloaded by you as a .pem file when you create the key pair.

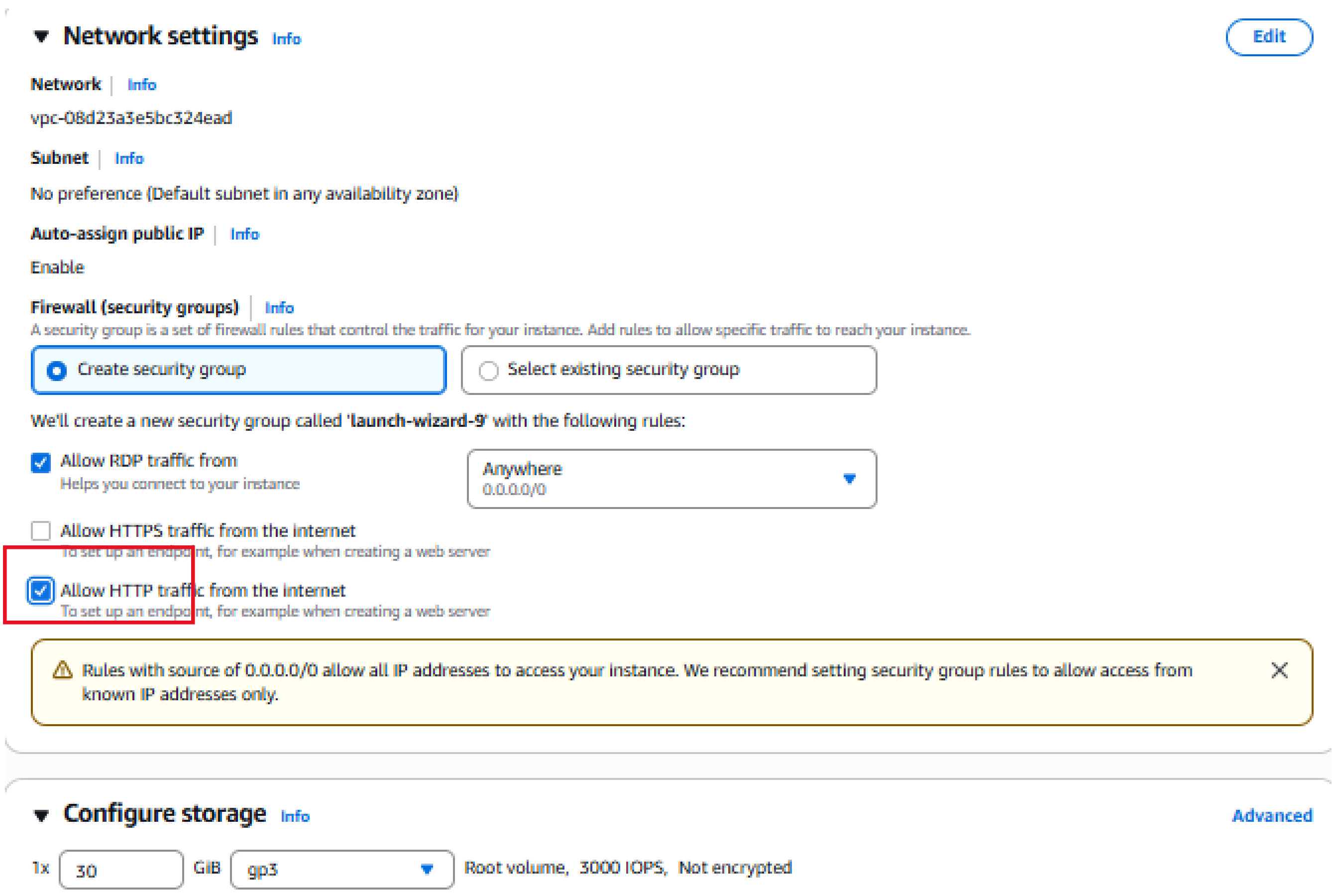
- When you connect via SSH (Linux) or RDP (Windows), AWS checks that your private key matches the public key stored on the instance.
- The public key is injected into your EC2 instance at launch:

  For **Linux instances**, it's placed inside the file: .ssh/authorized_keys.
  Use command: **ls -la** -> To see the hidden files.

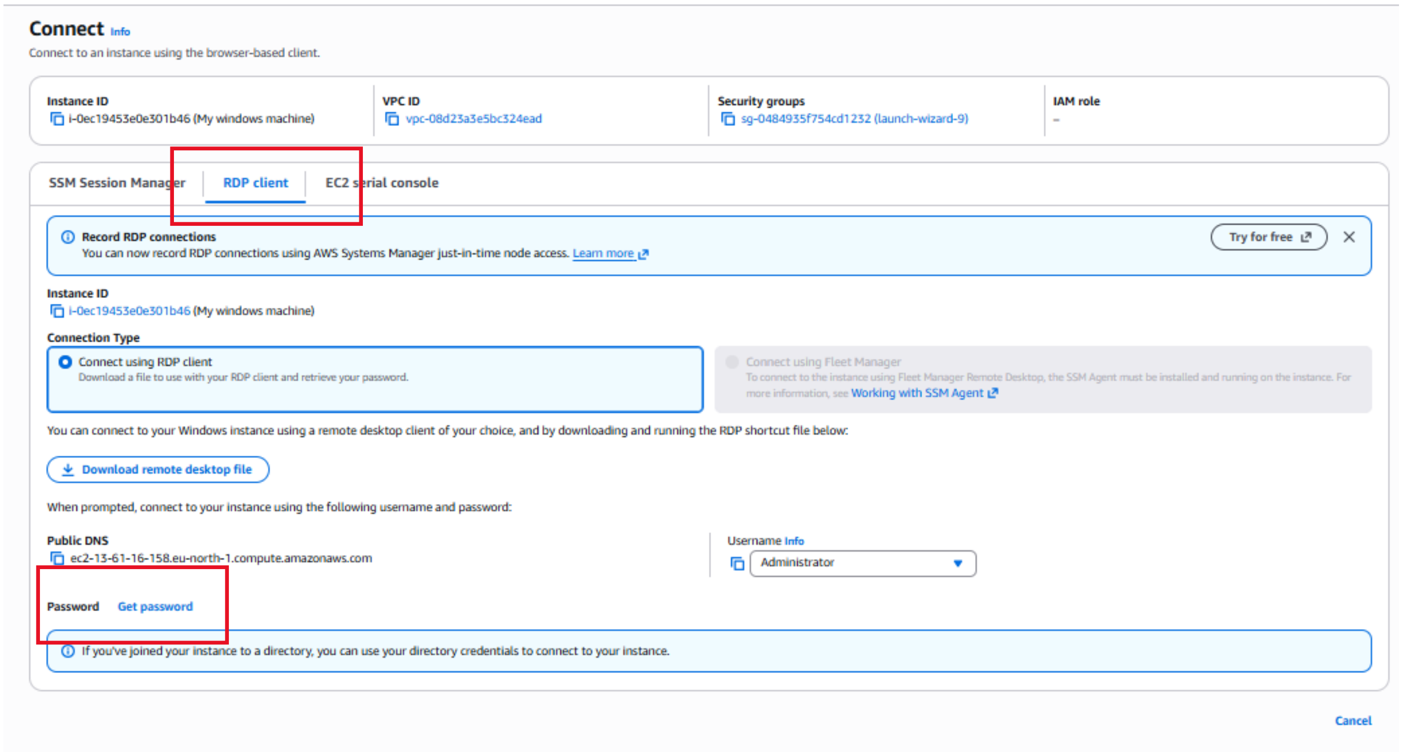
Hands on Web hosting using windows EC2 instance:

Step no:1 Create a Windows Ec2 instance and enable Https in the Network Settings. Launch instance.

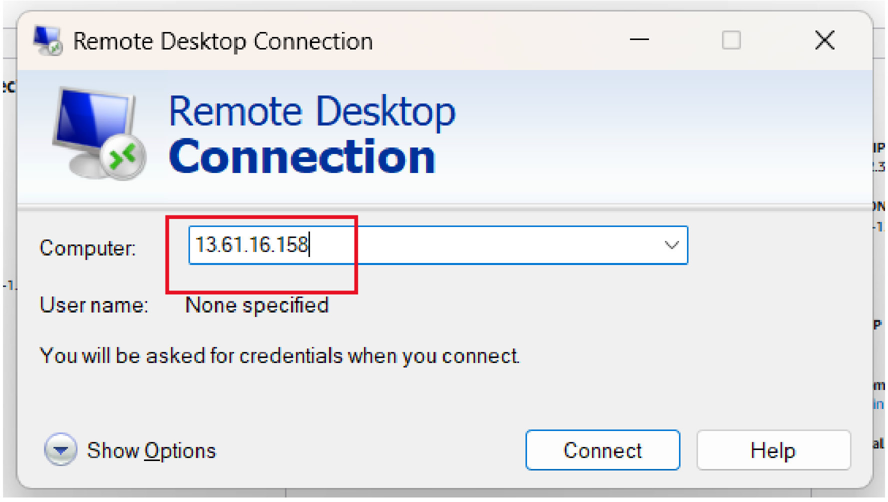




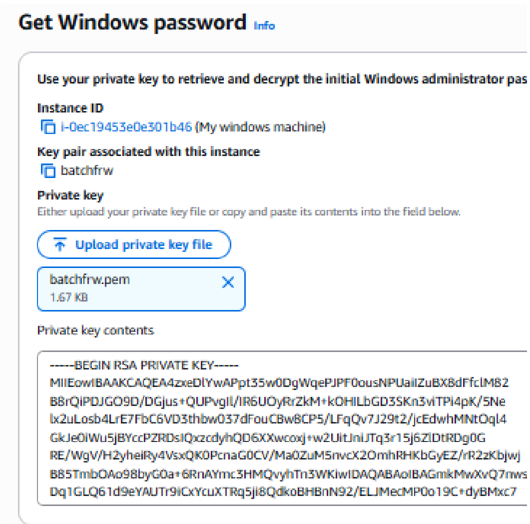
<u>Step no:2</u> Connect the windows instance. Go to the RDP client.

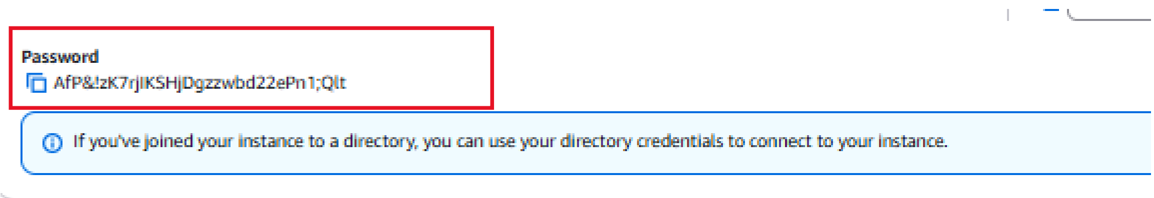


<u>Step no:3</u> Open Remote desktop connection in our local machine and open. Pate the public ip address of the ec2 instance. Click connect.

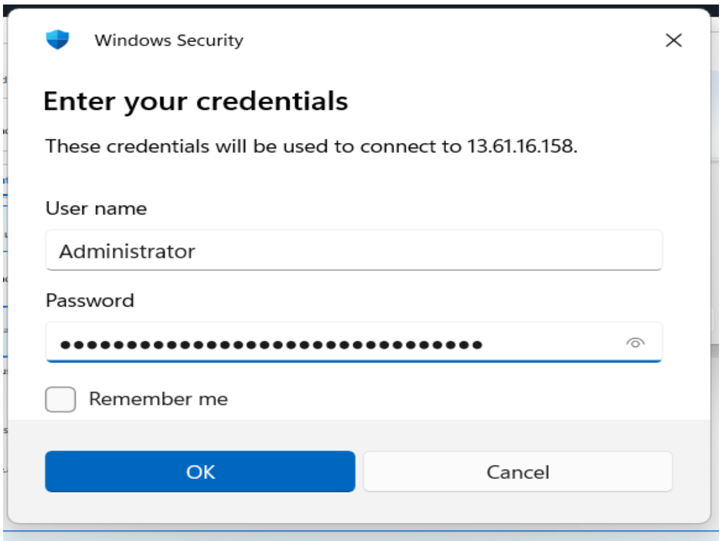


Get the password by uploading the keypair file and decrypt it.

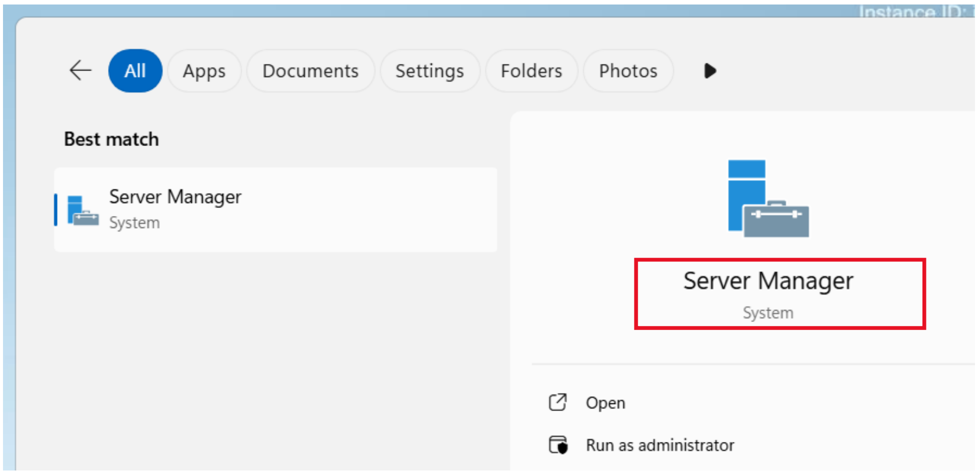




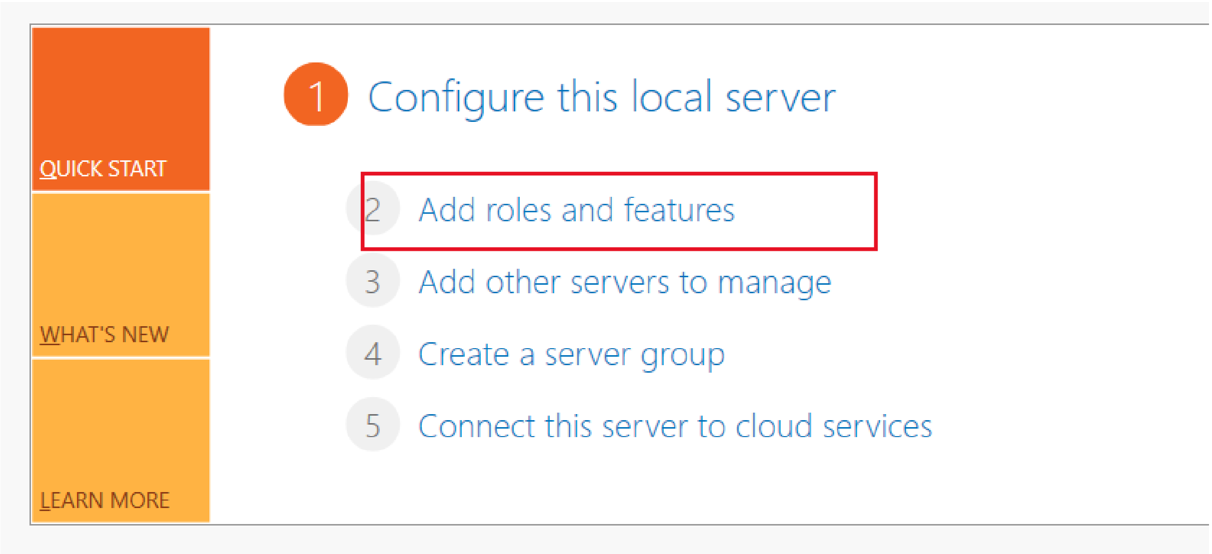
Enter the password in the RDC and click connect.

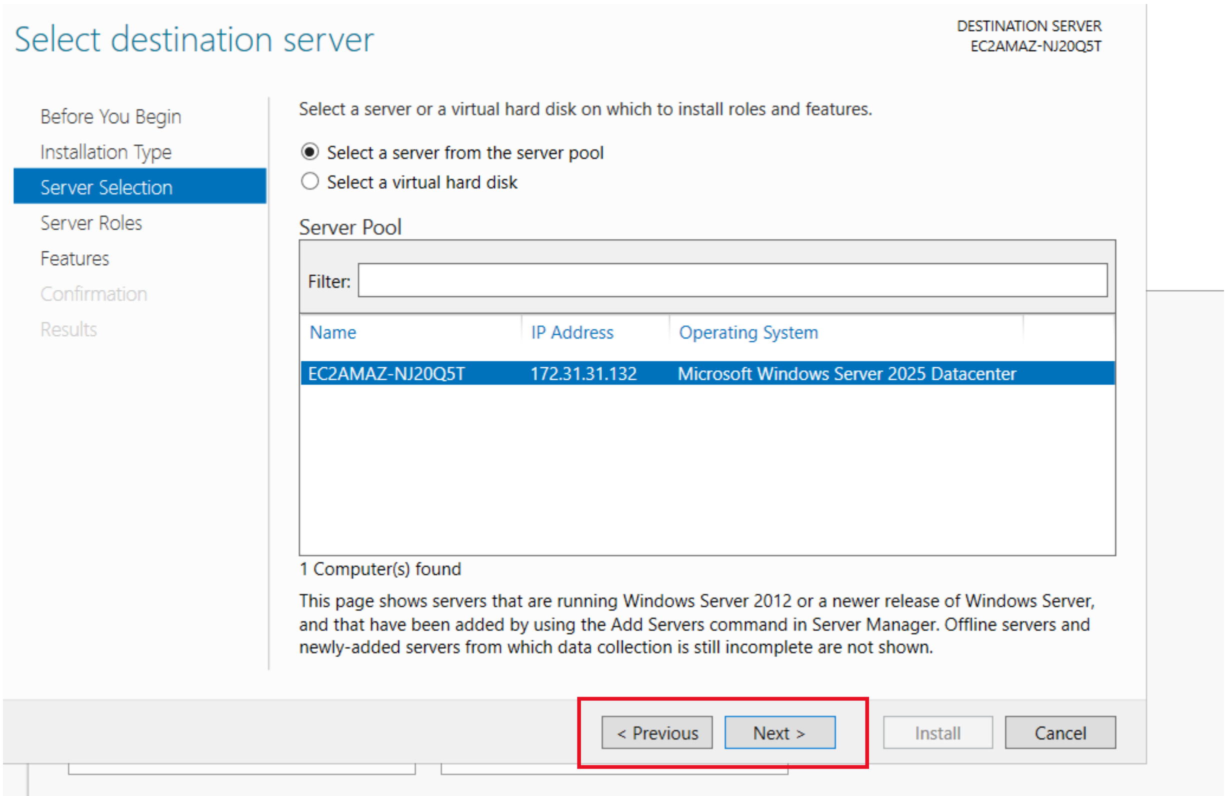


Step no:4 Now the windows machine will open. Search server manager and open it.

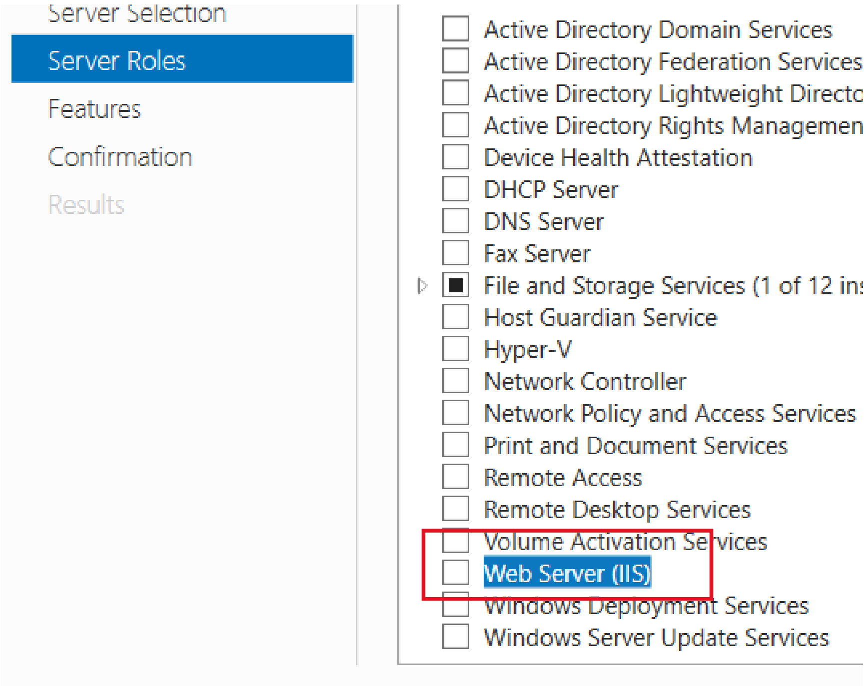


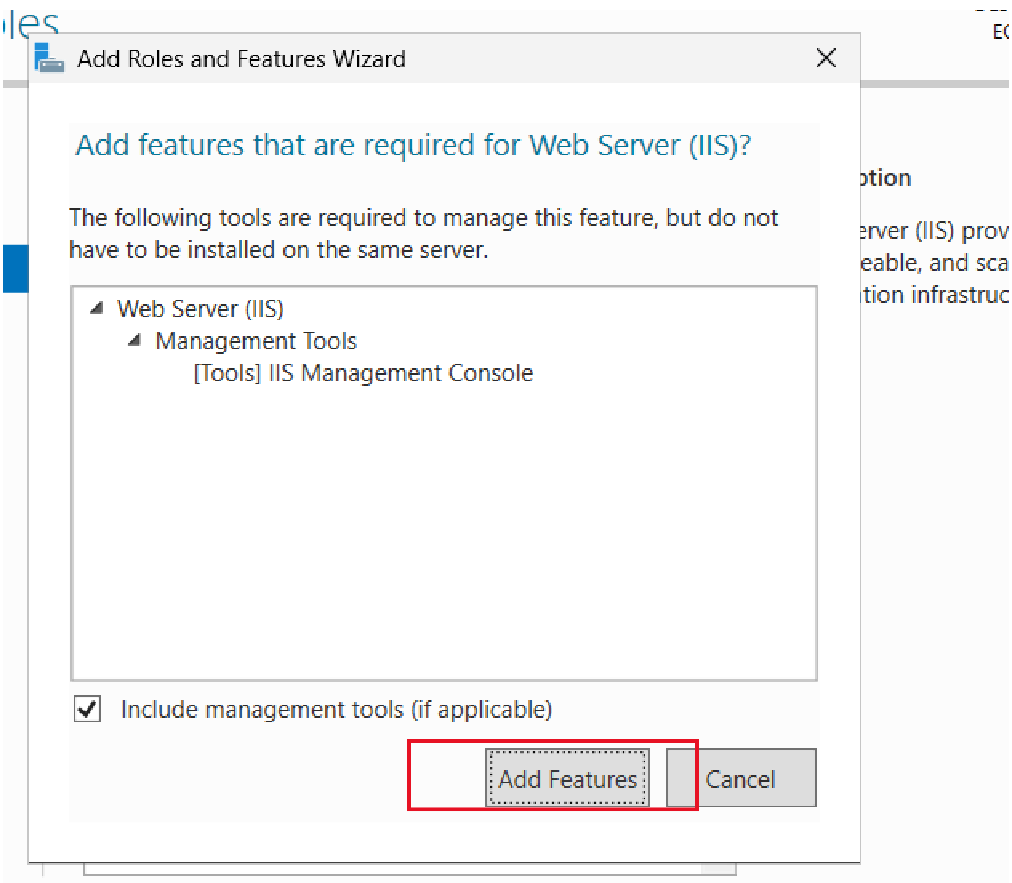
Step no:5 Click on add roles and features. Click on next until server roles.



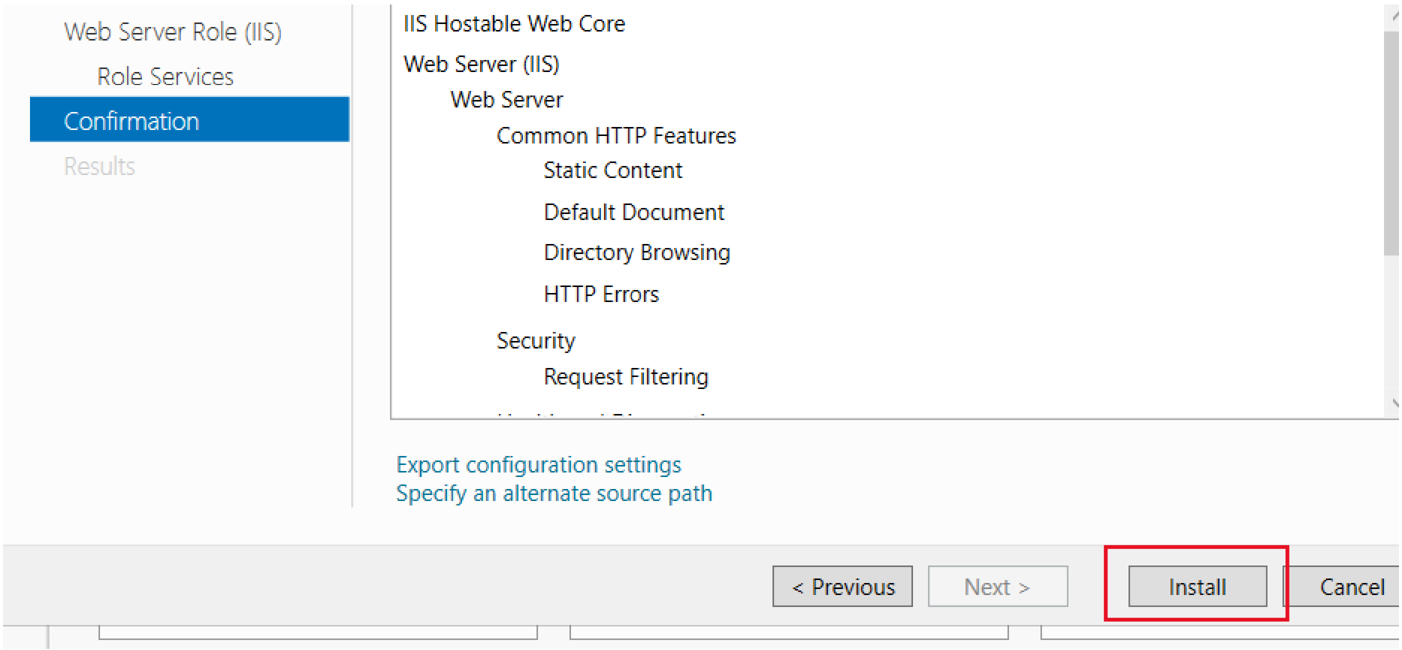


Step no:6 Click we server (IIS) and choose add features.

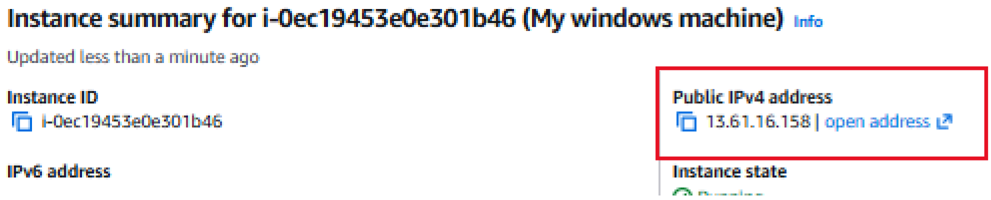


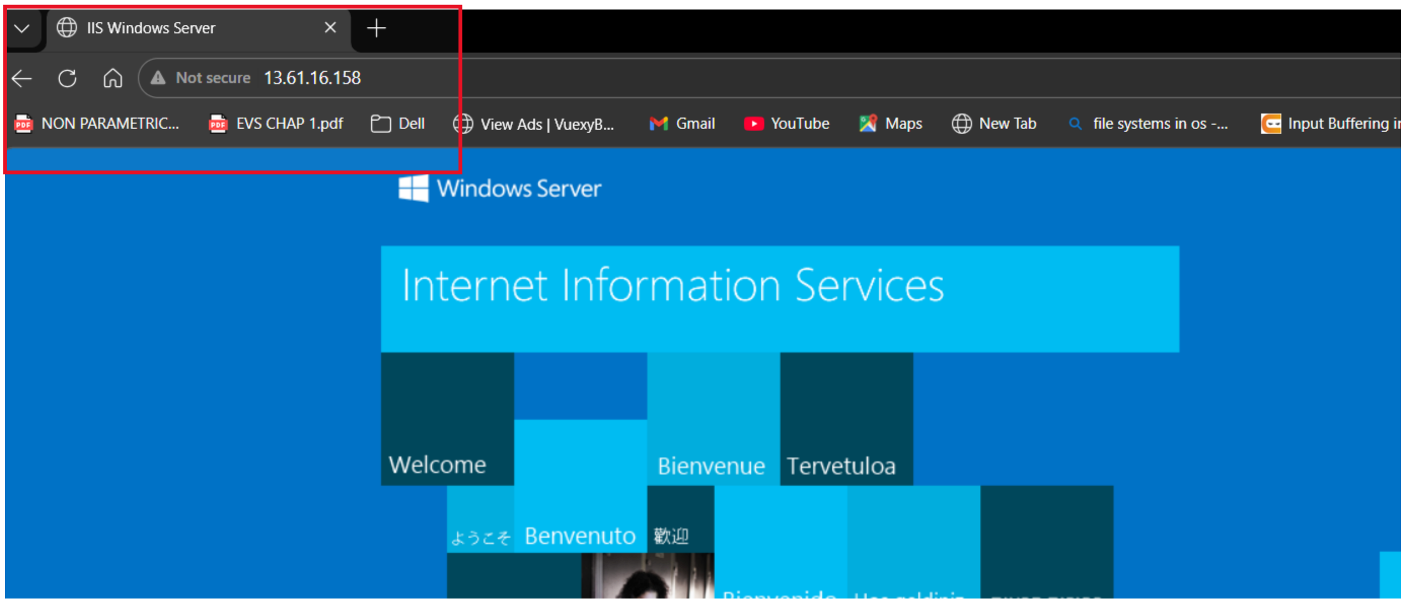


Step no:7 Now click install.

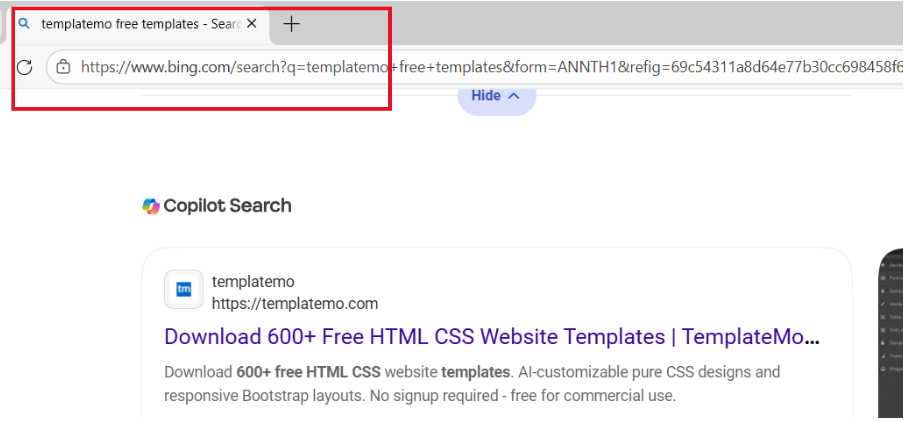


<u>Step no:8</u> Now copy the public ip address of the ec2 instance and paste it in an incognito browser we can see the output.

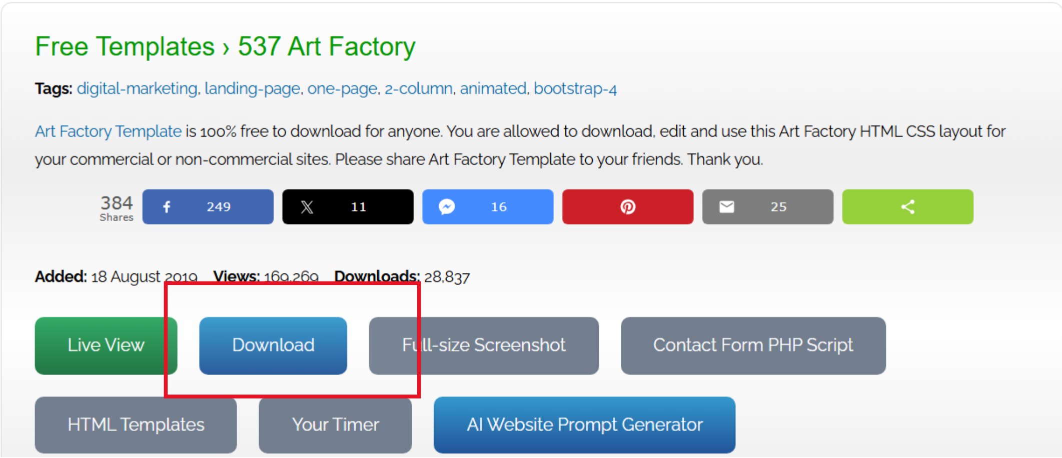




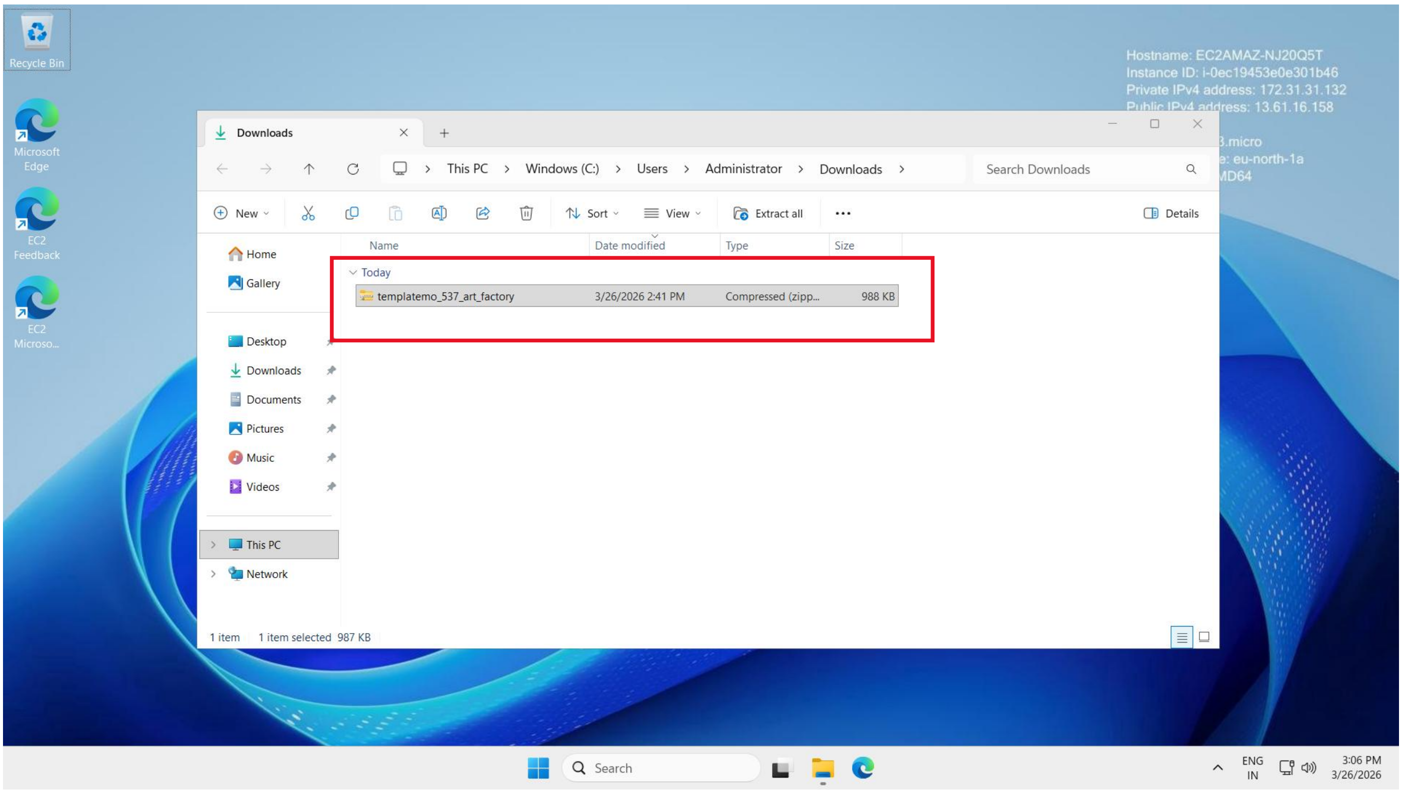
<u>Step no:9</u> Now we can add our file and we can host it. Go to our Windows instance. Open the edge browser , search for free templates.



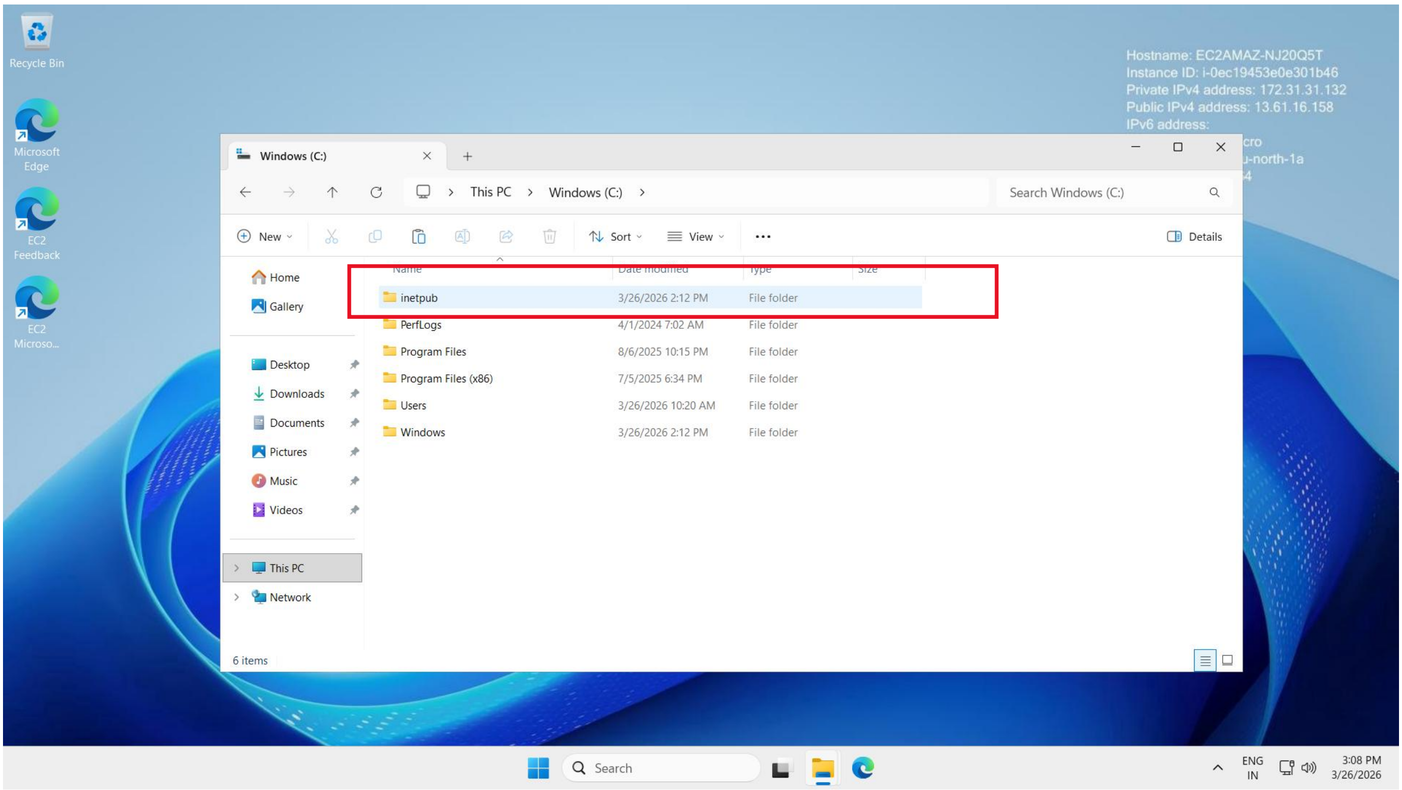
Step no:10 Now open it download it

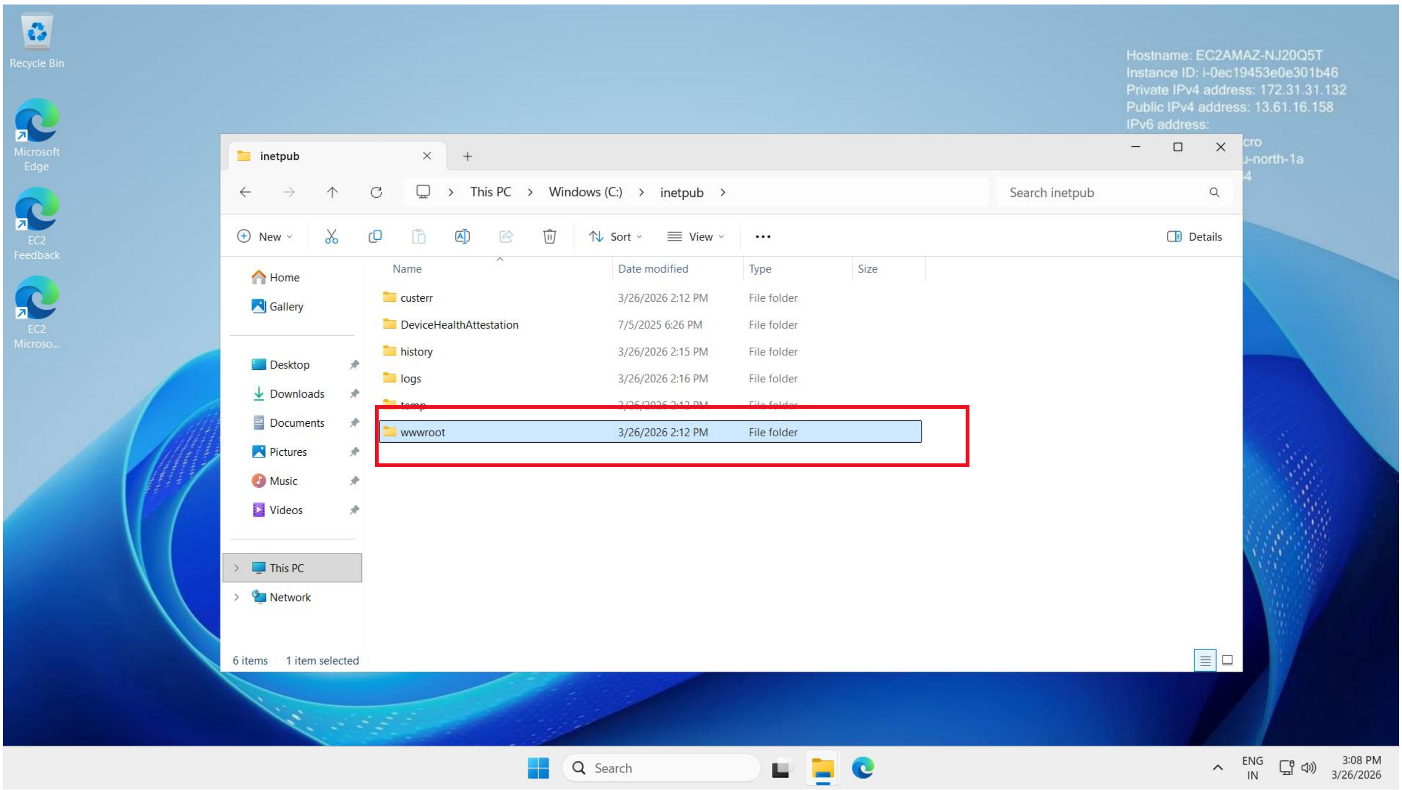


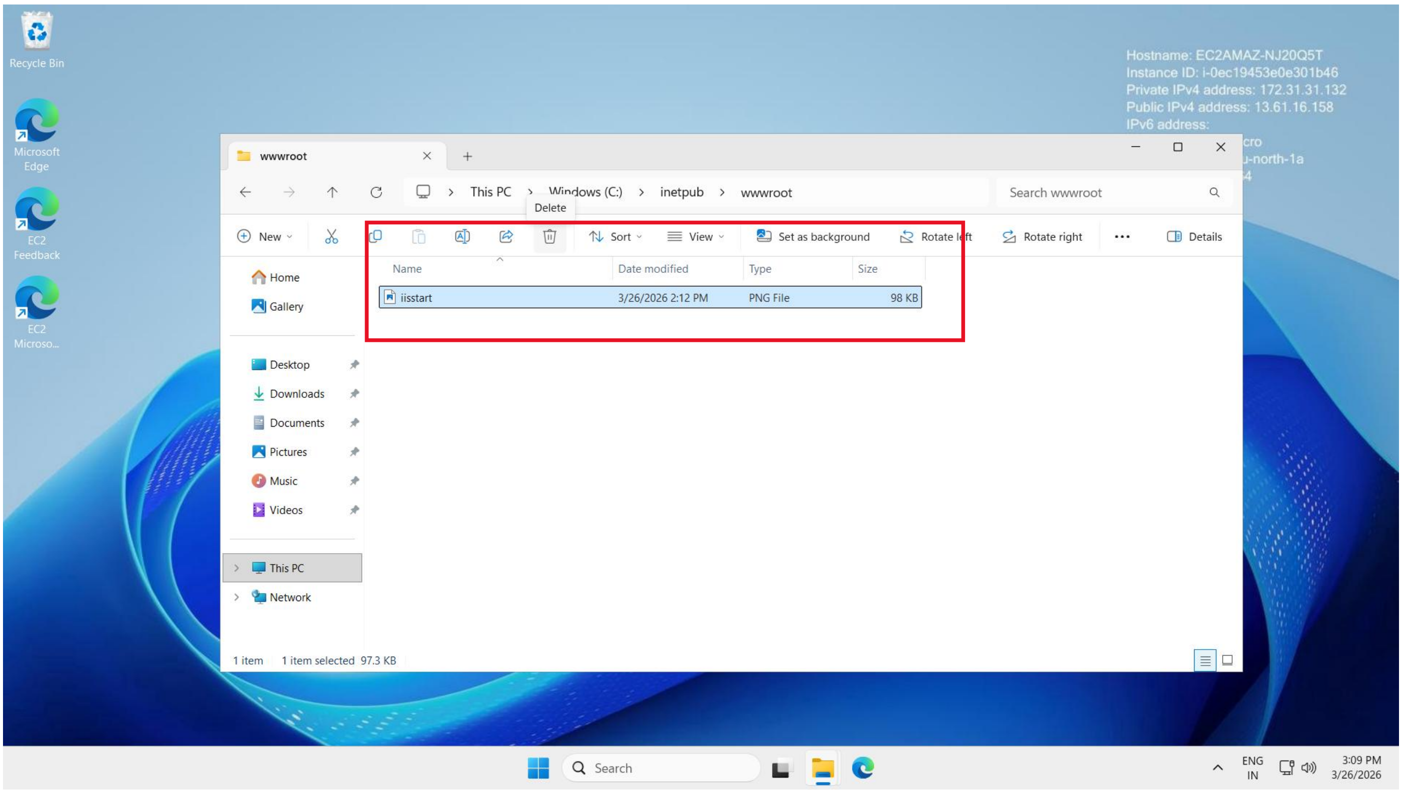
Step no:11 Now open the file in the downloads. It will be a zipfile. Extract it

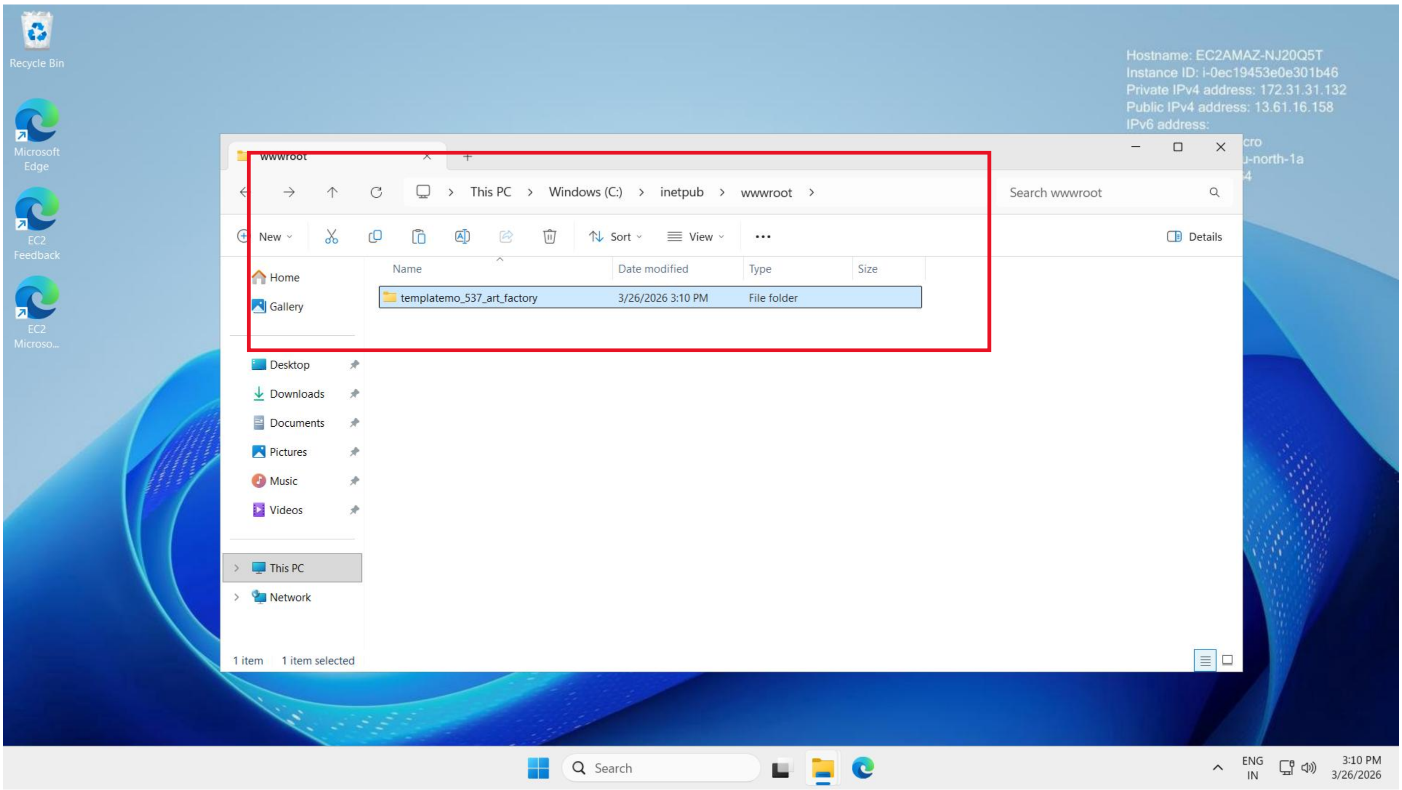


Step no:12 Now follow the path →this pc→ inetpub → wwwroot → delete the existing html file and paste our downloaded file.


Windows (C:)
This PC > Windows (C:) >
Search Windows (C:)
Name
Date modified
Type
Size
inetpub 3/26/2026 2:12 PM File folder
PerfLogs 4/1/2024 7:02 AM File folder
Program Files 8/6/2025 10:15 PM File folder
Program Files (x86) 7/5/2025 6:34 PM File folder
Users 3/26/2026 10:20 AM File folder
Windows 3/26/2026 2:12 PM File folder
6 items
Hostname: EC2AMAZ-NJ20Q5T
Instance ID: i-0ec19453e0e301b46
Private IPv4 address: 172.31.31.132
Public IPv4 address: 13.61.16.158
IPv6 address:


inetpub
This PC > Windows (C:) > inetpub >
Search inetpub
Name
Date modified
Type
Size
custerr 3/26/2026 2:12 PM File folder
DeviceHealthAttestation 7/5/2025 6:26 PM File folder
history 3/26/2026 2:15 PM File folder
logs 3/26/2026 2:16 PM File folder
temp
wwwroot 3/26/2026 2:12 PM File folder
6 items 1 item selected
Hostname: EC2AMAZ-NJ20Q5T
Instance ID: i-0ec19453e0e301b46
Private IPv4 address: 172.31.31.132
Public IPv4 address: 13.61.16.158
IPv6 address:





Now copy the public ip of the ec2 and paste it in a private browser. We can see the output of our code.